

Companion XXXIV: Filaments, Coherence, and the Geometry of the Cosmic Web in the Relaxation-Driven Cyclic Cosmology

Michael Lehmann

`mi.lehmann@gmx.de`

April 2026

Abstract

The cosmic web is the most striking manifestation of non-linear structure formation. Its filaments, sheets, and voids encode the dynamical history of the Universe and reflect the coherence of the underlying gravitational field. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential alters the geometry and dynamics of the web in a characteristic way. This Companion develops a continuous description of filament formation, examines the influence of the relaxation scale on web coherence, and derives analytical expressions for filament thickness, curvature, and stability. The resulting picture reveals a cosmic web that is smoother, more coherent, and more strongly shaped by large-scale modes than in the standard cosmological model.

1 Introduction

The cosmic web emerges from the anisotropic collapse of matter under gravity. Filaments form where collapse proceeds along two principal axes, sheets where only one axis collapses, and voids where expansion dominates. In the standard cosmological model, the geometry of the web is governed by the interplay between the initial tidal field and the non-linear amplification of density perturbations.

In RDCC, the relaxation-modified potential evolution introduces a new layer of structure. Large-scale modes retain a higher degree of temporal coherence, while small-scale modes experience a suppressed evolution. This imbalance reshapes the web's architecture: filaments become smoother, voids expand more coherently, and the connectivity of the web reflects the influence of the relaxation scale k_{rel} .

2 Filament Formation in RDCC

Filaments arise from the collapse of matter along two eigenvectors of the tidal tensor. The collapse time is determined by the evolution of the gravitational potential $\Phi(k, a)$, which in RDCC takes the form

$$\Phi_{\text{RDCC}}(k, a) = \Phi(k, a) \left[1 - \chi_{\Phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (1)$$

Modes with $k \ll k_{\text{rel}}$ evolve coherently, while modes with $k \gg k_{\text{rel}}$ experience a reduced potential decay. This difference modifies the collapse sequence. Filaments form earlier in regions dominated by large-scale coherence, and their cross-sectional profiles become smoother because the suppressed small-scale modes contribute less to the steepening of the density gradient.

The characteristic filament radius can be estimated by balancing the tidal compression with the relaxation-modified potential:

$$r_{\text{fil}} \sim \left[\frac{\Phi_{\text{RDCC}}}{\lambda_1 - \lambda_2} \right]^{1/2}, \quad (2)$$

where λ_1 and λ_2 are the dominant eigenvalues of the tidal tensor.

3 Coherence and Curvature of Filaments

The coherence of a filament refers to the degree to which its spine follows a smooth trajectory through space. In RDCC, the enhanced temporal stability of large-scale modes leads to a more coherent gravitational field, which in turn reduces the curvature of filament spines.

The curvature κ of a filament can be approximated by

$$\kappa \sim \frac{|\nabla_{\perp} \Phi_{\text{RDCC}}|}{|\nabla_{\parallel} \Phi_{\text{RDCC}}|^2}, \quad (3)$$

where the subscripts denote derivatives perpendicular and parallel to the filament axis. Because the relaxation mechanism suppresses small-scale fluctuations, the perpendicular gradient becomes smoother, reducing κ and producing straighter, more extended filaments.

This effect is particularly pronounced in regions where the tidal field is dominated by modes near k_{rel} , which act as a stabilizing backbone for the web.

4 Void Expansion and Web Connectivity

Voids expand as matter flows outward along the eigenvectors of the tidal tensor. In RDCC, the suppression of small-scale potential evolution reduces the formation of substructure within voids, allowing them to expand more coherently. The void radius evolves approximately as

$$R_{\text{void}}(a) \propto a^{\gamma_{\text{RDCC}}}, \quad (4)$$

where γ_{RDCC} exceeds its standard-cosmology counterpart due to the enhanced influence of large-scale modes.

The connectivity of the web is also altered. Filaments linking massive halos become more stable, while the number of small-scale filaments decreases. This produces a web with fewer but more prominent connections, reflecting the imprint of the relaxation scale on the geometry of structure formation.

5 Observational Signatures

The RDCC modifications to the cosmic web manifest in several measurable ways. Weak-lensing maps reveal smoother filamentary structures and a reduced abundance of small-scale peaks. Galaxy surveys show enhanced alignment of galaxies along filaments, reflecting the increased coherence of the underlying tidal field. Void statistics exhibit larger effective radii and a narrower distribution of shapes, consistent with the suppression of small-scale perturbations.

These signatures provide a direct observational window into the relaxation scale and the temporal structure of the gravitational field in RDCC.

6 Conclusion

The cosmic web in the Relaxation-Driven Cyclic Cosmology is shaped by the scale-dependent evolution of the gravitational potential. Filaments form earlier and with smoother profiles, voids

expand more coherently, and the connectivity of the web reflects the influence of the relaxation scale. These features distinguish RDCC from the standard cosmological model and offer a rich set of observational tests. The present Companion establishes the theoretical foundation for future studies of galaxy alignment, filament lensing, and the topology of the cosmic web within the RDCC framework.

References

- [1] Lehmann, M. (2026). Relaxation-Driven Cyclic Cosmology (RDCC): A Minimal CPT-Symmetric Framework with a Single Parameter. Flagship Paper. Zenodo: <https://zenodo.org/records/19744958> (v43.0 or later).
- [2] The complete RDCC companion series (I–LVII) is available at the same Zenodo record above.
- [3] Companion XVI: Boltzmann Code and Modified Hierarchy.
- [4] Companion XXXI: RDCC Halo Model and Non-Linear Structure.
- [5] Companion XXXII–XXXIX: Galaxy–Halo Connection, Cosmic Web, Lensing, RSD, ISW, tSZ, Rotation Curves, CGM.
- [6] Planck Collaboration (2020), *Astron. Astrophys.* 641, A6.
- [7] DESI Collaboration (2024/2025), arXiv: [aktuelle $f\sigma_8$ /BAO-Paper].
- [8] Maiolino et al. (2024), *Nature* (JWST high- z galaxies/quasars).
- [9] Kaiser, N. (1987), *Mon. Not. R. Astron. Soc.* 227, 1 (RSD).
- [10] Bartelmann, M. & Schneider, P. (2001), *Phys. Rep.* 340, 291 (Weak Lensing Review).
- [11] Hu, W. & Okamoto, T. (2002), *Astrophys. J.* 574, 566 (CMB Lensing).
- [12] Lewis, A. & Challinor, A. (2006), *Phys. Rep.* 429, 1 (CMB Lensing Review).
- [13] Navarro, J. F., Frenk, C. S. & White, S. D. M. (1997), *Astrophys. J.* 490, 493 (Halo Profiles).